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ME10 Final Project
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Ballon D'Or (Pewter Ball)



The Ballon D'Or is a trophy that is awarded each year to the soccer player regarded as the best in the world in a given calendar year. Many of the greatest soccer players in the world have been given this award including Messi and Ronaldo. For our ME10 final project we decided to create this award out of pewter. The project went through several iterations - we were originally planning on sand casting two halves of the ball and then assembling them and mounting the finished ball on some sort of stand. One of our group members suggested an alternative casting approach - lost wax casting using a wax mold made from a vacuum molder mold. It was therefore decided to make one-half of the ball using sand casting and one half of the ball using lost wax casting. Both procedures, however, proved to be much more difficult than originally anticipated. The ball was to be cast with a core inside of it in order to hollow out the ball, thus reducing the weight of the ball and the amount of pewter used to create the ball. Creating a core proved to be the ultimate challenge and at the end of the day the final casting solution was neither sand casting nor lost wax casting – instead we created a plaster mold and core using an actual half of a soccer ball and poured the pewter into that mold to create the part. The final part was then attached to bolts using more pewter and mounted to a piece of plywood. To finish off the piece, a CNC router was used to write the artists' names in the wood. We would like to thank the professors and the staff at the Bray Machine Shop for all of their help on this project.